

Team description paper

BagelScience

Our robot is an autonomous soccer robot designed for the RoboCupJunior Soccer competition. The robot is built to operate quickly and accurately in a fast-paced environment while making real-time decisions during gameplay. Using a combination of infrared sensors, camera vision, and orientation tracking, the robot can locate the ball, identify goals, and navigate the field without human control. The system is powered by a Teensy 4.1 microcontroller, which allows fast processing and smooth communication between all components.

The robot uses multiple infrared sensors to detect the ball and maintain awareness of its surroundings. An OpenMV camera is integrated for colour and goal detection, enabling the robot to identify the correct target and react dynamically during matches. A Bosch BNO055 gyroscope helps maintain stable orientation so the robot always knows the direction of play and can recover quickly after collisions or sudden movements.

What sets our robot apart is the combination of advanced sensing systems, fast response time, and efficient software integration. The robot is designed for both speed and precision, allowing it to react quickly while maintaining controlled movement and accurate targeting. Its modular electronic system and optimised sensor arrangement improve reliability and performance compared to many standard competition robots.

Team

Yotam. This is my third year in RoboCup, and the second year I am competing in an international championship. I worked on the 3D module design and the IR sensors programming.

Yaniv: I worked on the design of our PCB. Last year, I competed in the European soccer infrared category.

Itay Malka: I worked on our PCB design, the camera module, and the robot's general main code. This is my third year in RoboCup and my second time at the European championship.

Itay Rlpin: I am competing in the European championship for the second time. I've worked on the 3D CAD design and the component picking.

Project planning

Overall Project Plan

Objective: The primary objective of team bagelscience is to develop a fast and precise autonomous soccer robot capable of 360-degree ball detection, optimal orbiting maneuvers, and efficient scoring actions while maintaining highly accurate field positioning.

Requirements & Constraints:

Mechanical: Implementation of a 4-motor omnidirectional drivetrain to allow seamless vector movement regardless of the chassis orientation.

Technical: High-speed data processing from a peripheral IR array and integration with an OpenMV camera for real-time visual anchor detection.

Performance: Low-latency response times to ensure sensor data continuity and stability during high-speed movements.

Key Milestones:

Drivetrain & Chassis (M1): Design and manufacturing of the robot chassis and omni-gear assemblies to ensure a stable mechanical foundation.

Sensor Array (M2): Integration of 24 IR sensors via multiplexers (Mux) to achieve precise distance and angle readings from the ball.

Computer Vision (M3): Calibration of the OpenMV camera for robust color and shape detection to calculate goal distances and angles.

Control & Localization (M4): Development of a heading-stabilization algorithm using the BNO055 IMU and creation of a Python GUI for real-time spatial monitoring.

Strategy Logic (M5): Implementation of scoring strategies and autonomous recovery mechanisms (Stall detection).

Integration Plan

The system architecture is centered around a Teensy 4.1 microcontroller, which coordinates all sensor data:

Ball Detection: A ring of 24 IR sensors is scanned via multiplexers, providing the central controller with high-resolution ball coordinates.

Vision System: An OpenMV H7 Plus camera processes visual data and transmits target coordinates via I2C/UART for real-time trajectory correction.

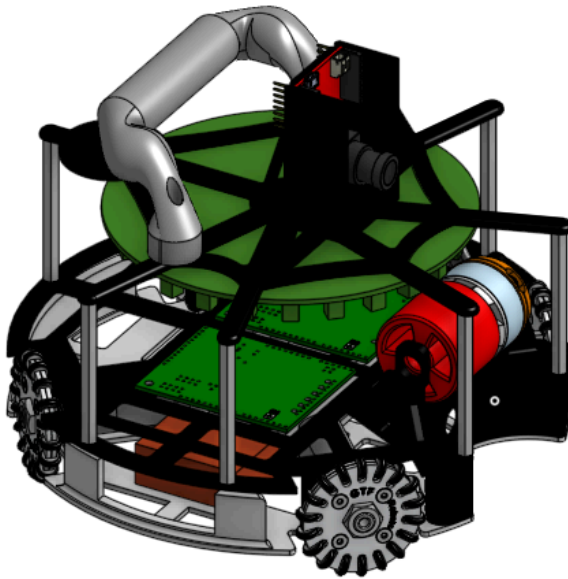
Orientation & Stability: A BNO055 IMU provides absolute Yaw data, allowing the robot to maintain a fixed field orientation through a proportional (P-loop) control system.

Hardware

High-Level Hardware Overview

Our RoboCupJunior Soccer robot was designed for fast movement, accurate ball tracking, stable orientation, and reliable autonomous gameplay. The robot combines a lightweight mechanical structure with advanced electronics to react quickly and precisely during matches. The system integrates infrared ball sensors, an OpenMV camera for goal detection, a gyroscope for orientation tracking, and a Teensy 4.1 microcontroller for real-time processing and motor control. A custom PCB connects and organizes all electronic systems efficiently. Key features include 360-degree ball detection, omnidirectional movement, stable orientation tracking, modular layered construction, and compact internal organization.

Mechanical Design and Manufacturing



The robot's mechanical structure was designed to be lightweight, compact, and durable while protecting internal electronics during collisions. The chassis uses a low center of gravity to improve stability during rapid movement and turning. Components were carefully positioned to distribute weight evenly and improve balance during gameplay. The structure was built according to RoboCup size requirements while also allowing easy repairs, upgrades, and maintenance access.

The robot uses high-speed DC motors connected to omni wheels, allowing movement in all directions without needing to rotate first. This omnidirectional drivetrain improves maneuverability, reaction speed, and positioning during offensive and defensive gameplay. Special motor mounts were designed to reduce vibration and maintain wheel alignment during impacts. A front dribbler mechanism was added to improve ball control, maintain possession while moving, and increase shooting accuracy. The dribbler was positioned carefully to avoid blocking the sensors while still maximizing control of the ball.

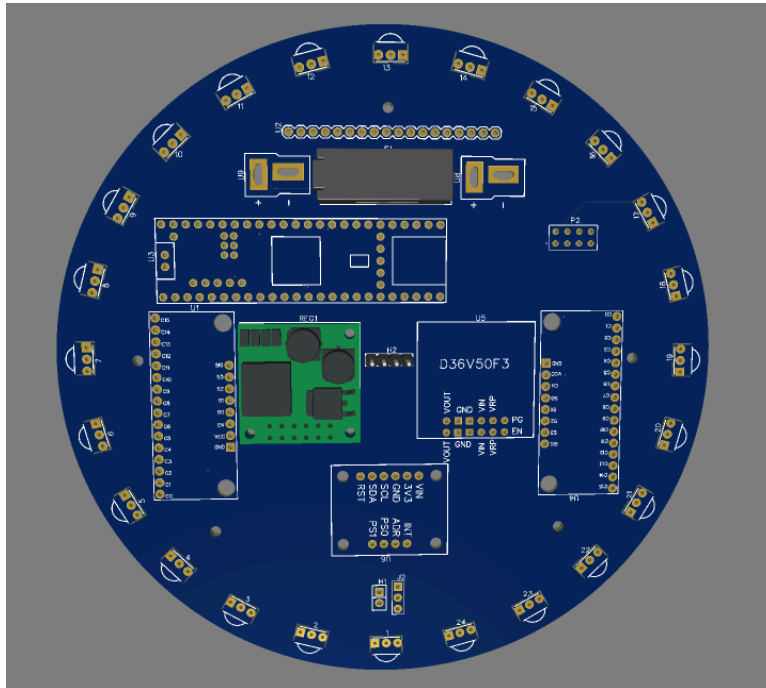
The robot was designed using CAD software before manufacturing to optimize component placement, weight distribution, and internal space usage. Multiple prototype versions were tested before selecting the final design. The final robot was assembled using modular sections, allowing damaged components to be replaced quickly during testing and competition. The robot also uses a layered design with separate drivetrain, electronics, and sensor sections, improving cable organization and simplifying future upgrades.

Mechanical testing included movement testing, collision testing, dribbler testing, and sensor placement evaluation. The drivetrain was tested for speed, turning accuracy, and stability during rapid direction changes. Collision tests ensured the frame could withstand repeated impacts without damaging internal systems. Dribbler tests focused on ball control, possession during turns, and shooting consistency. Different sensor positions were also

tested to maximize ball detection accuracy and goal visibility. The final design demonstrated stable movement, reliable ball control, strong structural durability, and accurate sensor coverage during gameplay.

One innovative aspect of the design is the compact omnidirectional drivetrain combined with optimized weight distribution, allowing fast and stable movement. Another important feature is the modular layered structure, which simplifies maintenance and allows easier upgrades while keeping the robot compact and competition-ready.

Electronic Design and Manufacturing



The robot's electronic system was designed for fast processing, reliable sensor integration, and stable power distribution. The main controller is the Teensy 4.1, selected because of its high processing speed, compact size, and ability to manage multiple communication protocols simultaneously. The Teensy processes all sensor data, controls motor drivers, communicates with the camera and gyroscope, and executes the robot's autonomous gameplay algorithms in real time.

The robot uses 24 TSSP4038 infrared sensors mounted around the robot to provide near 360-degree ball detection. These sensors detect the infrared light emitted by the soccer ball and determine its direction and distance relative to the robot. Because the large number of sensors exceeded the available controller input pins, two CD74HC4067 multiplexers were used to organize sensor readings while reducing wiring complexity and improving PCB layout efficiency.

An OpenMV Cam H7 camera was added for goal detection and colour recognition. The camera communicates with the Teensy through UART or I2C communication and improves gameplay by allowing accurate goal targeting. To maintain orientation during gameplay, the robot uses a Bosch BNO055 gyroscope and orientation sensor. This sensor continuously

tracks the robot's heading and helps maintain accurate movement even after collisions or wheel slipping.

The power subsystem uses Pololu D36V50F5 and Pololu D36V50F3 voltage regulators to provide stable operating voltages for the sensors, camera, controller, and logic systems. Stable voltage regulation prevents communication errors, electrical noise, and controller resets during gameplay. Dedicated motor drivers were also included to provide smooth acceleration, accurate speed control, and safe handling of motor current loads.

A custom PCB was designed to integrate the multiplexers, voltage regulators, sensor connections, communication lines, and power distribution into a compact and reliable system. The PCB reduced wiring complexity, improved reliability, minimized electrical noise, and simplified maintenance. It was also designed to fit within the robot's size limitations while supporting modular upgrades and clean sensor communication.

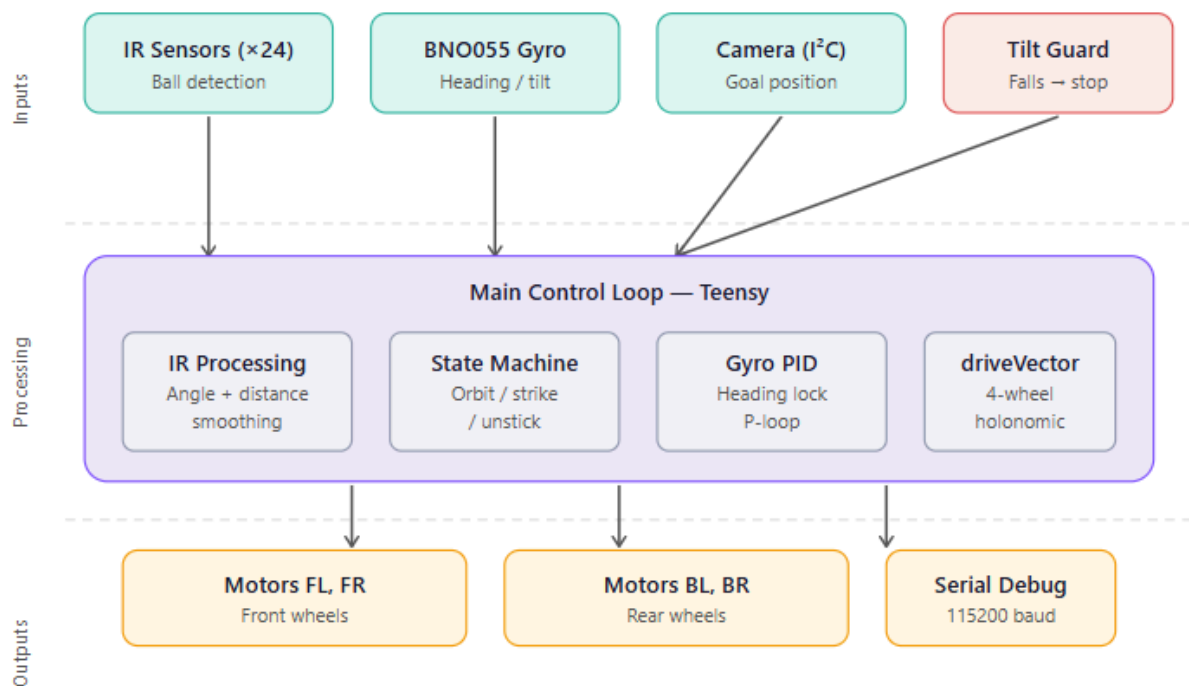
Electronic testing included individual infrared sensor testing, gyroscope calibration, camera testing under different lighting conditions, and full system gameplay simulations. The infrared sensor array successfully detected the ball from nearly all directions with fast response times and reliable readings. The gyroscope maintained accurate orientation during rapid movement and collisions, while the camera consistently detected goals during gameplay. Full system tests included ball chasing, goal targeting, defensive movement, and collision recovery, demonstrating reliable autonomous operation and stable sensor integration.

One innovative feature of the electronic system is the high-density infrared sensor array combined with multiplexer-based expansion, allowing accurate ball tracking without excessive hardware complexity. Another important innovation is the integration of infrared tracking, camera-based goal detection, and orientation sensing into a single coordinated control system, improving both offensive and defensive gameplay performance.

In conclusion, the hardware design focused on speed, stability, reliability, and autonomous gameplay. Through the combination of a durable mechanical structure, omnidirectional drivetrain, advanced sensor systems, stable power management, and high-speed processing, the robot achieved strong performance suitable for competitive RoboCupJunior Soccer matches.

Software

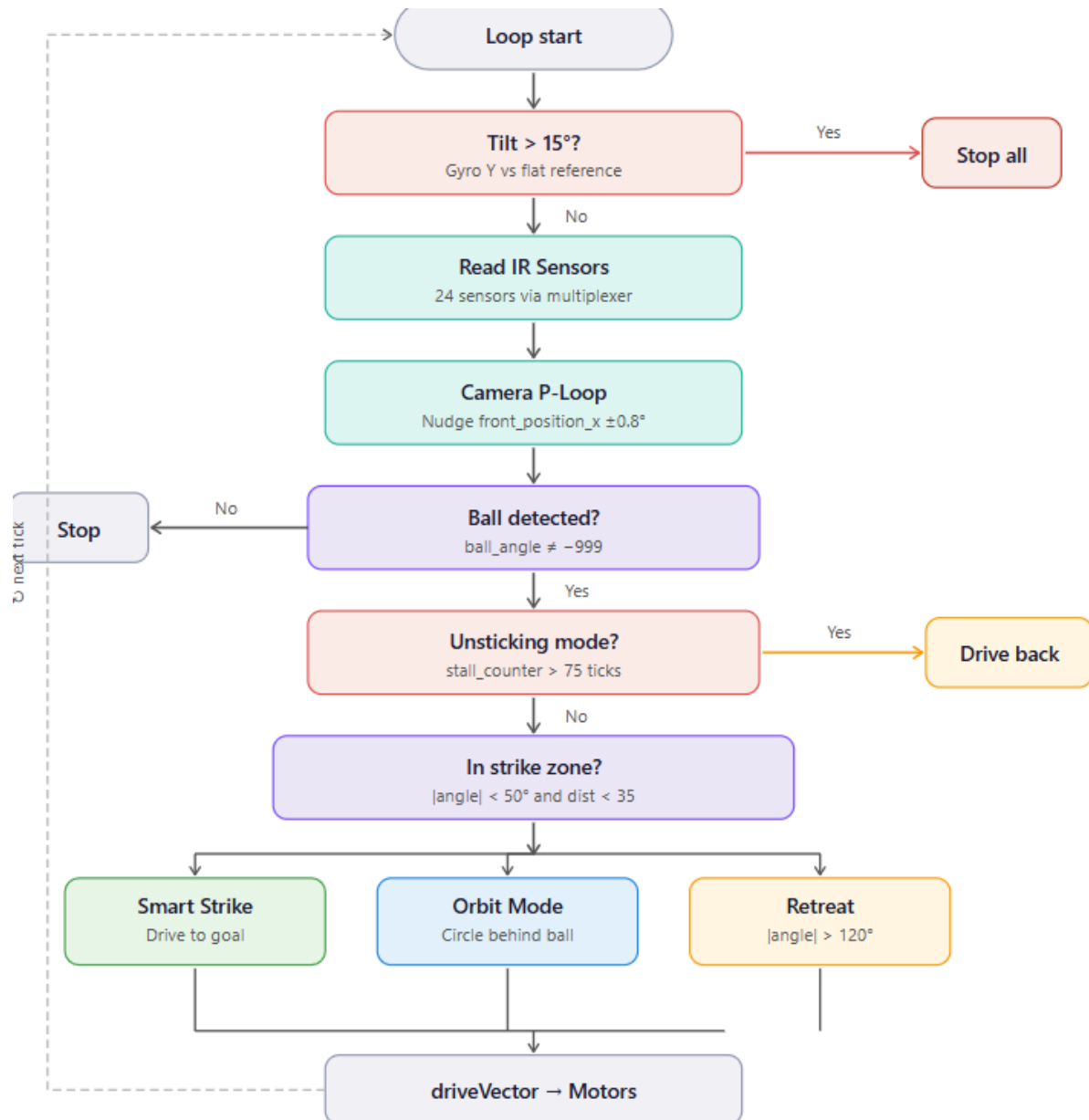
General Software Architecture



The robot's software runs on a microcontroller in a tight real-time loop of approximately one millisecond per tick. Every iteration reads all sensors, evaluates a tactical state machine, computes a holonomic drive vector, and writes PWM signals to the four motors. There is no operating system or task scheduler - the loop is intentionally bare-metal to minimize latency between sensor events and motor response.

The software is organized into three functional layers. The input layer consists of 24 IR sensors, a BNO055 nine-axis gyroscope, and a camera module that communicates over I²C. These feed a central processing layer running on the microcontroller, which handles all sensing, decision-making, and drive computation. The output layer is the four-motor holonomic drivetrain, with serial debug output running in parallel at 115,200 baud during development. The camera communicates via an I²C interrupt handler (`receiveEvent`), meaning goal data arrives asynchronously and is merged into the main loop on each tick rather than blocking it.

Main control loop:



Each tick of the main loop follows a fixed sequence. First, the gyroscope Y-axis is read and compared against a calibrated flat-surface reference stored at startup. If the tilt exceeds 15°, all motors stop immediately, and the function returns — this is a hard safety gate that cannot be bypassed by any other logic. Second, all 24 IR sensors are polled via two 16-channel analog multiplexers. The sensor with the longest low-pulse duration is identified as the strongest IR signal, its array index is mapped to a bearing angle in degrees, and that angle is smoothed with an exponential filter to reduce single-tick noise spikes. Third, a proportional correction is applied to the stored heading target based on the latest camera command, gradually steering the robot's facing toward the opponent's goal. Finally, the tactical state machine is evaluated, and the resulting angle and speed are passed to the drive function.

Tactical state machine

At any moment, the robot is in one of five mutually exclusive movement states. In the idle state, no IR signal is detected, and all motors stop. In the retreat state, the ball is more than 120° from dead-ahead, so the robot drives straight backward to avoid inadvertently

pushing the ball into its own goal. In the orbit state, the ball is between 50° and 120° off-center or beyond the strike distance threshold, and the robot executes a gain-scaled circular arc around the back of the ball. In the smart strike state, the ball is within 50° of dead-ahead and within the distance threshold, so the robot drives directly toward the goal while staying locked onto the ball. Finally, the unstick state is triggered when the ball angle has not changed meaningfully for 75 consecutive ticks, indicating the robot is physically jammed; it drives backward for 250 ms and then re-enters normal operation.

Holonomic drive

All movement is computed inside a single drive function. A proportional loop on the gyroscope heading generates a rotation correction term. The desired movement angle is decomposed into sine and cosine components for a 45° -offset four-wheel mecanum configuration, and the rotation term is added or subtracted to each motor to maintain heading while translating in any direction. All four outputs are then normalized so no motor exceeds the maximum PWM value of 255.

Libraries and tools

The software relies on the Adafruit BNO055 library for gyroscope communication, the Arduino Wire and Wire1 two-wire libraries (with Wire1 dedicated to the gyro on separate hardware pins to avoid bus contention with the camera), a custom Motor class that wraps direction and PWM into a single call, and a custom Gyro wrapper that handles angle-wrapping arithmetic so heading error always stays within the -180° to $+180^\circ$ domain. IR sensors are sampled using `pulseIn()` in low-timeout mode so the 24-sensor scan completes without blocking the loop for a meaningful duration.

Integration problems solved in software.

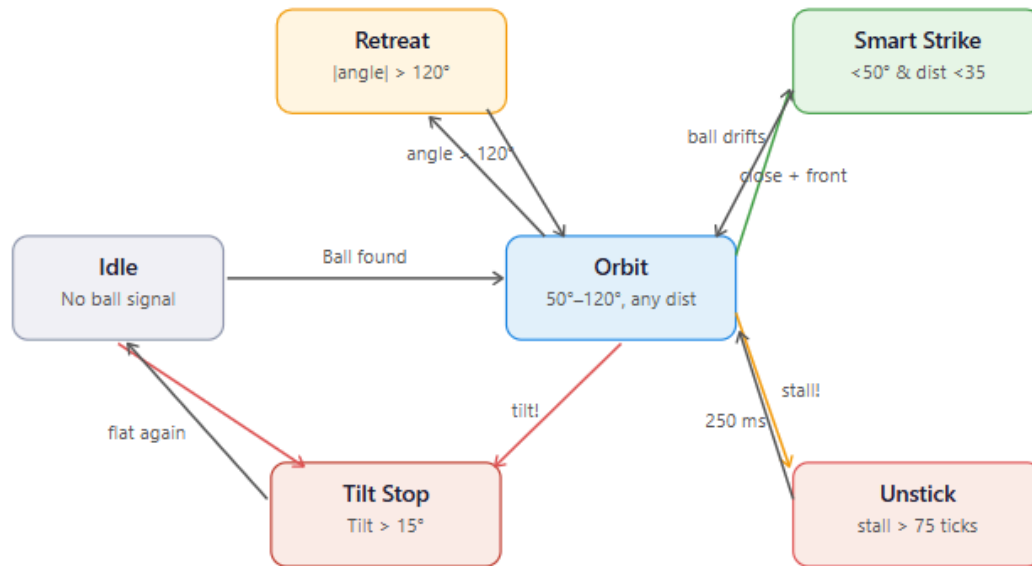
Gyro drift during rotation was addressed by storing the heading as an absolute reference set at startup and continuously correcting it with a proportional loop, so the robot always knows which direction is "forward" regardless of how many rotations it has made.

IR sensor noise was handled with an exponential smoothing filter with a coefficient of 0.4, which weights the most recent reading at 40% and accumulated history at 60%. This damps single-frame spikes while still reacting quickly to genuine ball movement.

I²C bus conflict between the camera and the gyro was resolved by placing them on two separate hardware I²C peripherals, eliminating address collisions without any software arbitration overhead.

Camera latency was addressed by using an interrupt handler that caches the latest goal command and a timestamp whenever a new byte arrives. The main loop reads from this cache and can check whether the data is fresh, preventing the robot from acting on outdated goal information after it has already re-oriented.

Innovative Solutions



Smart strike with goal correction:

The most significant algorithmic innovation is the blended strike approach. A naive implementation would drive straight at the ball; a goal-aware one would steer purely toward the opponent's goal. The problem with the second approach is that if the ball and the goal are not perfectly aligned, the robot loses contact with the ball mid-approach and the shot fails. The solution blends 80% ball tracking with 20% goal correction: the robot stays physically in contact with the ball throughout the drive while still steering the shot toward the goal. The correction direction is derived from the camera — when the goal is to the right, a negative offset is applied, when it is to the left, a positive offset is applied. When the robot is already very close to the goal, it stops rather than overshooting, preventing rebounds off the back wall.

Adaptive orbit gain

Rather than a fixed circular orbit radius, the orbit gain changes based on both which side the ball is on and how close it is. Four distinct gain values were used — higher when close and on the left, lower when far and on the right. This asymmetry compensates for a systematic mechanical bias in the drivetrain and ensures the robot consistently arrives at the back of the ball rather than the front, making the transition into the strike state smooth and repeatable.

Stall detection and automatic recovery

The robot monitors whether the ball's apparent angle is changing from tick to tick. If 75 or more consecutive ticks pass with a change smaller than 0.3° , the robot concludes it is physically jammed against the ball or the field wall. Instead of grinding indefinitely, it executes a 250 ms full-speed escape maneuver in reverse, resets the stall counter, and re-enters the normal state machine. This mechanism was validated by intentionally wedging the robot against the field wall and confirming that recovery occurred within one second in all trials.

Camera-driven heading nudge

Goal alignment is applied incrementally to the robot's stored heading target at $\pm 0.8^\circ$ per loop tick rather than as a discrete steering command. This means the robot gradually rotates to face the goal during the orbit phase, so by the time it enters the strike zone, it is already

correctly oriented rather than needing a sharp last-second correction that could destabilize the approach.

Testing procedures

Validation was conducted in three stages. In the first stage, each subsystem was tested in isolation: IR processing output was printed to serial and compared against manual measurements of ball position across a range of angles and distances, while the gyro proportional loop was tested by commanding fixed headings and recording settling time and steady-state error. In the second stage, state machine transitions were verified in the field by placing the ball at known positions and confirming that the correct state was entered and logged. In the third stage, full match simulations were run in which a human operator moved the ball around the robot, and the serial monitor was used to confirm that stall recovery, tilt protection, and goal correction all activated under the expected conditions. All tuning constants were finalized only after a minimum of 20 trials per configuration, with settling behavior, transition accuracy, and stall recovery latency recorded across each set.

Performance Evaluation

Evaluation against Challenges: The robot was evaluated on its ability to maintain trajectory accuracy and execute autonomous maneuvers consistently. A major challenge addressed was "linear drift" during movement; this was resolved by integrating IMU data into the vector-drive function, resulting in high stability during complex maneuvers.

Testing Procedures:

Digital Twin Visualization: We developed a custom Python GUI (`field_gui.py`) to visualize the robot's calculated position on a digital field. This allowed us to validate localization and vision algorithms visually before conducting physical field trials.

Sensor Calibration: Standardized calibration scripts (`calibration.py`) were used to determine the exact mathematical constants required for distance sensing across different lighting environments.

Recovery Mechanism Testing: We performed rigorous testing on the robot's ability to detect physical blockages using the `stall_counter` logic, confirming that the autonomous recovery routine successfully clears jams in under one second.

Analysis and Impact: Testing data indicated that smoother movement is achieved through the implementation of data smoothing parameters and dynamic motor speed control. The iterative testing process, which involved comparing serial logs to physical robot movement, led to significant optimizations in navigation logic and increased the overall reliability of the system.

Conclusion

Team BagelScience successfully designed and built a competitive autonomous soccer robot for the RoboCupJunior Soccer competition. By combining a robust omnidirectional drivetrain with a 24-sensor infrared array, OpenMV camera-based goal detection, and real-time gyroscope orientation tracking -all coordinated by a high-speed Teensy 4.1 microcontroller - the robot is capable of fast, intelligent, and reliable autonomous gameplay.

On the software side, key innovations such as the blended smart-strike algorithm, adaptive orbit gain, and automatic stall recovery demonstrate a thoughtful approach to solving real competition challenges. The system was thoroughly tested at both the component and full-match level, confirming stable performance across all major gameplay scenarios.